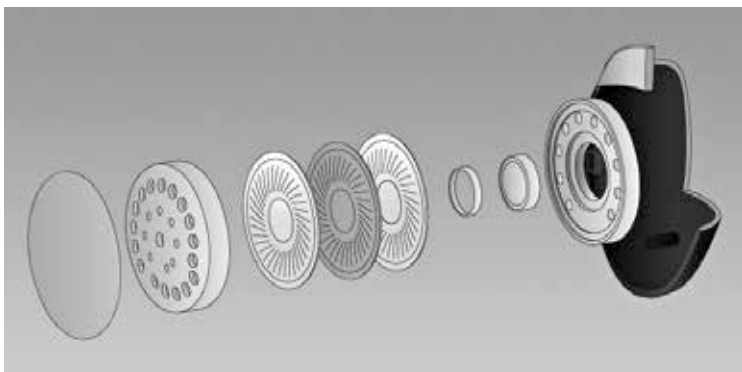


Innards of Headphones

Now that we are clear about the types, we can get into the inner workings of headphones. Headphones can be called as miniature speakers, which can be worn over the head. Headphones, like speakers carry drivers. A driver is the device which converts electrical signals into sound. Every driver has a magnet, voice coils and a diaphragm. When electrical signals are passed to it, the magnets are activated, which causes the diaphragm to vibrate. The way the drivers vibrate dictates the type of sound. By varying the signals to the driver, music or audio is made. Drivers are rated by their diameter. Each driver can be tuned to produce a certain type of sound. A driver, for example, can be tuned to produce bass heavy sounds, midrange and even as a tweeter. While most headphones are tuned to produce a flat sound, meaning without being heavy in sound types, some are tuned specifically



Structure of a Dynamic Driver manufactured by RHA

for the type of audio the listener prefers. The drivers used in headphones most commonly are referred to as dynamic drivers. There are some high end headphones which are able to accommodate multiple drivers, with audio signals being split and routed to the right driver.

The various types of drivers are explained below:

Dynamic Drivers

These use larger diaphragms, and produce sound without needing much power. These are preferred almost all regular headphones and are very common today. They do not focus much on the audio quality, instead deliver decent sound for a low cost.